

Revised Generation Interconnection System Impact Study Report

For Queue Project Q19-1162

Lamping Station 138 kV
September 01, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1162. The Interconnection Customer proposes a 184.6 MW solar photovoltaic facility interconnecting at the Lamping Station 138 kV substation in the Maryland area. The request was received on March 07, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 375 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$326,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Calvert Cliffs 345/138kV XFMR is overloaded to 106.6% of its 575 MVA emergency rating for the loss of the Cady – Easton 2 345.0 kV. The project contributes 7.9% DFAX (14.5 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Calvert Cliffs 345/138kV XFMR	Replace transformer with higher-rated unit	\$25,321,000	1% (\$326,000)
	Total allocated to project		\$326,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.